野火_F407霸天虎_原理图_V2.0

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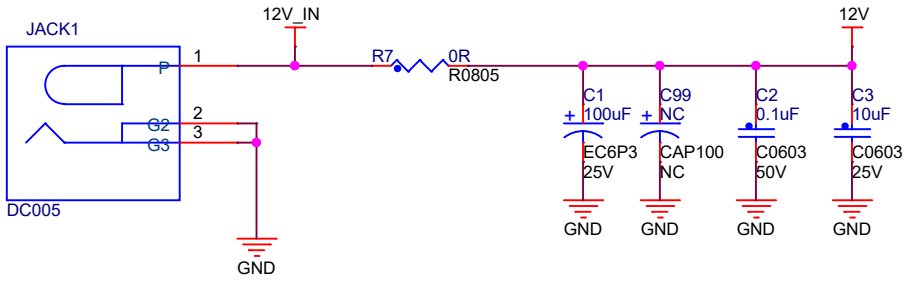
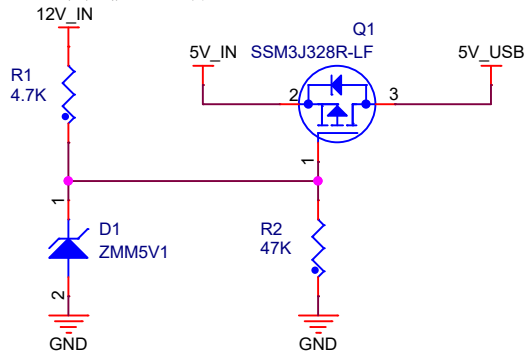
历史版本

版本号	日期	设计	描述
V1.0	2017-12-1	cancore	初始版本
V2.0	2018-12-22	cancore	更新结构，增加一个RS232串口，增加一个EBF Module接口
V2.1	2024-08-26	LPC	更新 CAN 485 232 的替代料

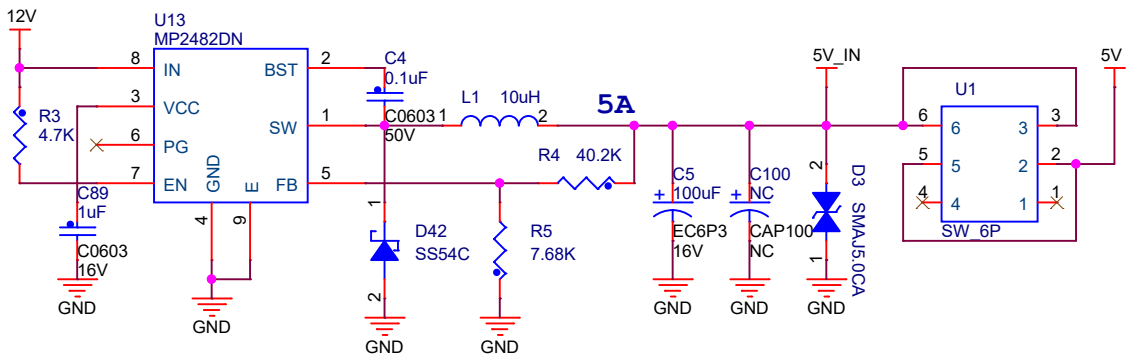
DC IN : 6V~12V



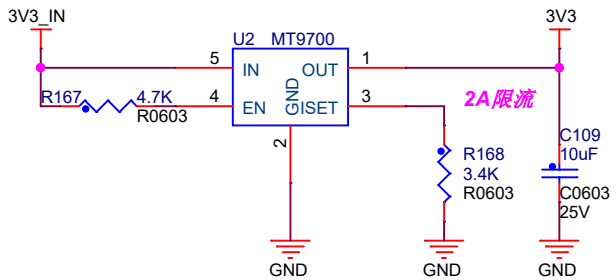
外部电源供电时断开USB供电，防止灌电流，保护电脑USB口



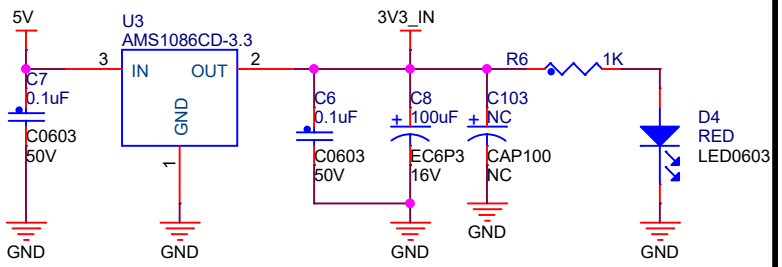
DC/DC 5V



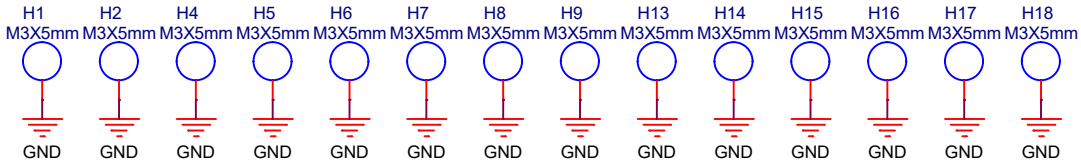
3.3V 输出过流保护



LDO 3.3V



定位孔



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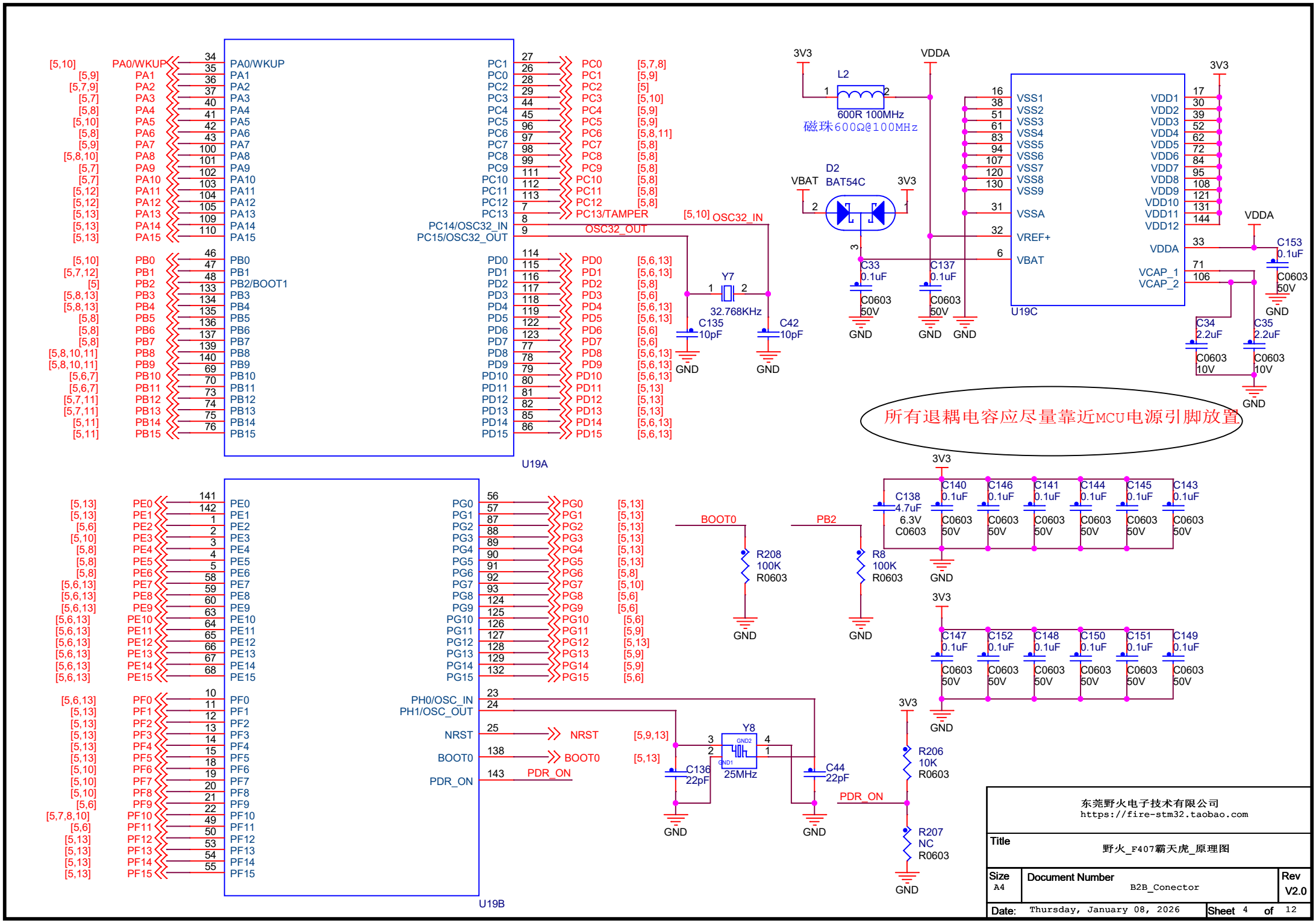
U19A
U19B
U19C
L2
600R 100MHz
磁珠600Ω@100MHz
D2
BAT54C
Y7
32.768KHz
Y8
25MHz
C135 10pF
C42 10pF
C136 22pF
C44 22pF
C33 0.1uF
C137 0.1uF
C153 0.1uF
C34 2.2uF
C35 2.2uF
C138 4.7uF 6.3V C0603
R208 100K R0603
R8 100K R0603
R206 10K R0603
R207 NC R0603
BOOT0
PB2
PDR_ON
NRST
OSC32_IN
OSC32_OUT
所有退耦电容应尽量靠近MCU电源引脚放置
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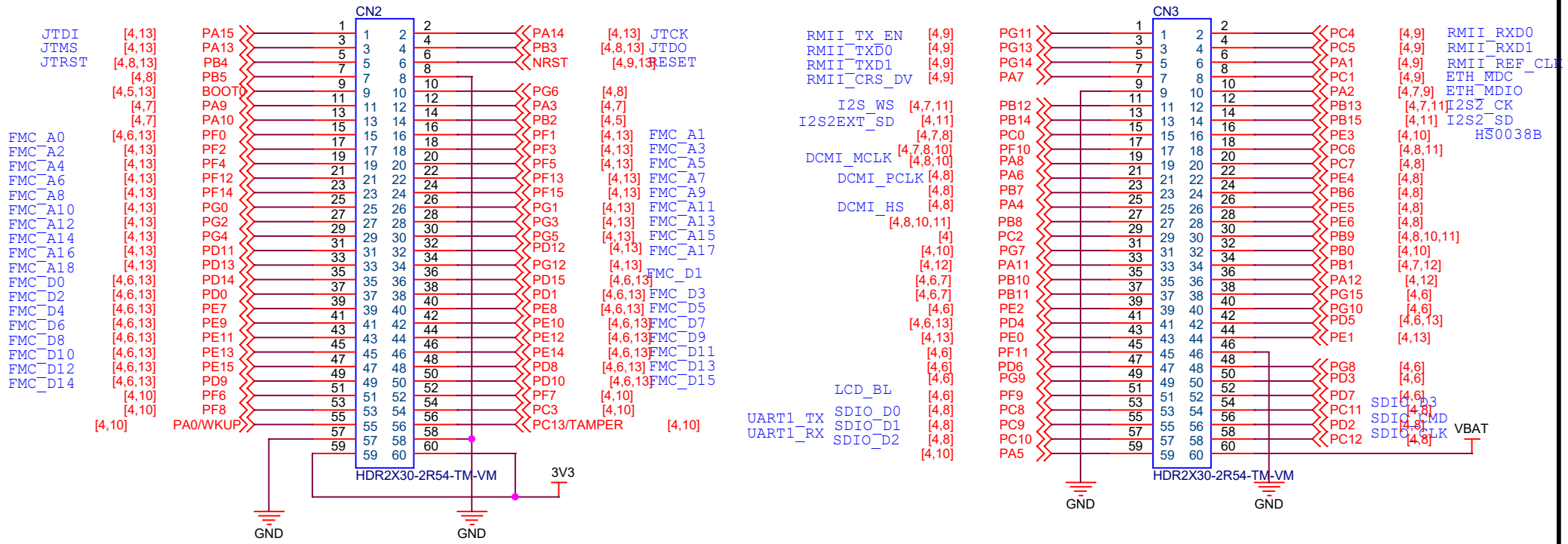
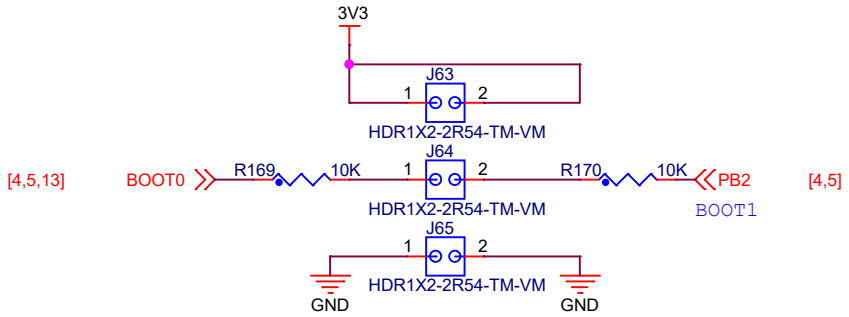
引出IO口

GPIO引出110个，除了晶振占用4个IO其他全部引出。

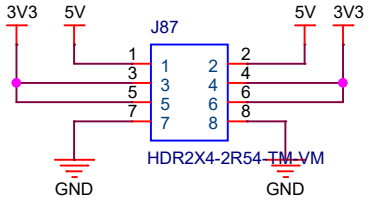


设置启动模式



F407 BOOT设置		
BOOT0	BOOT1	启动方式
0	X	用户闪存
1	0	系统存储器/ISP
1	1	内嵌SRAM
默认配置是用户闪存（即内部Flash）		

引出电源



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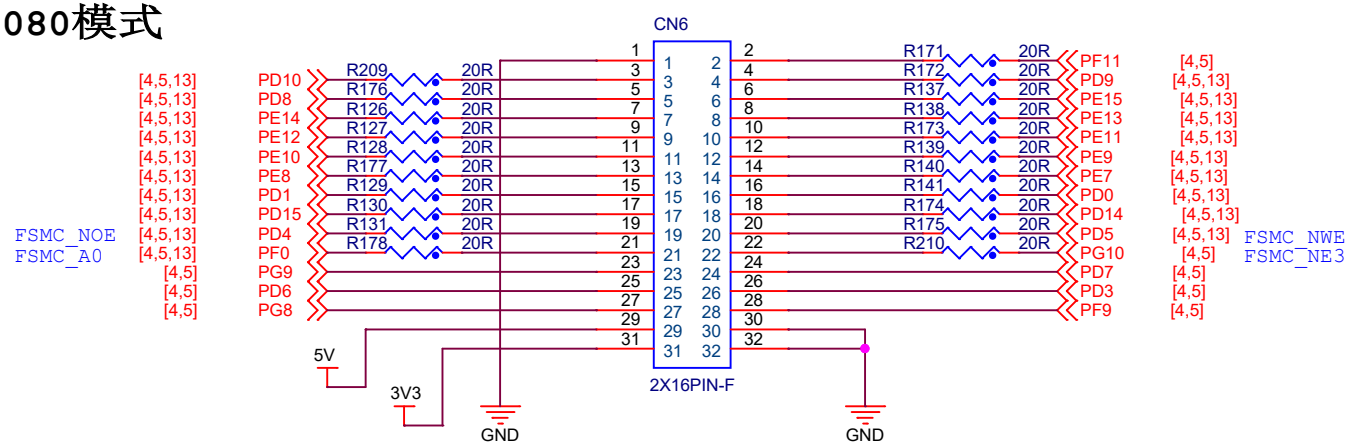
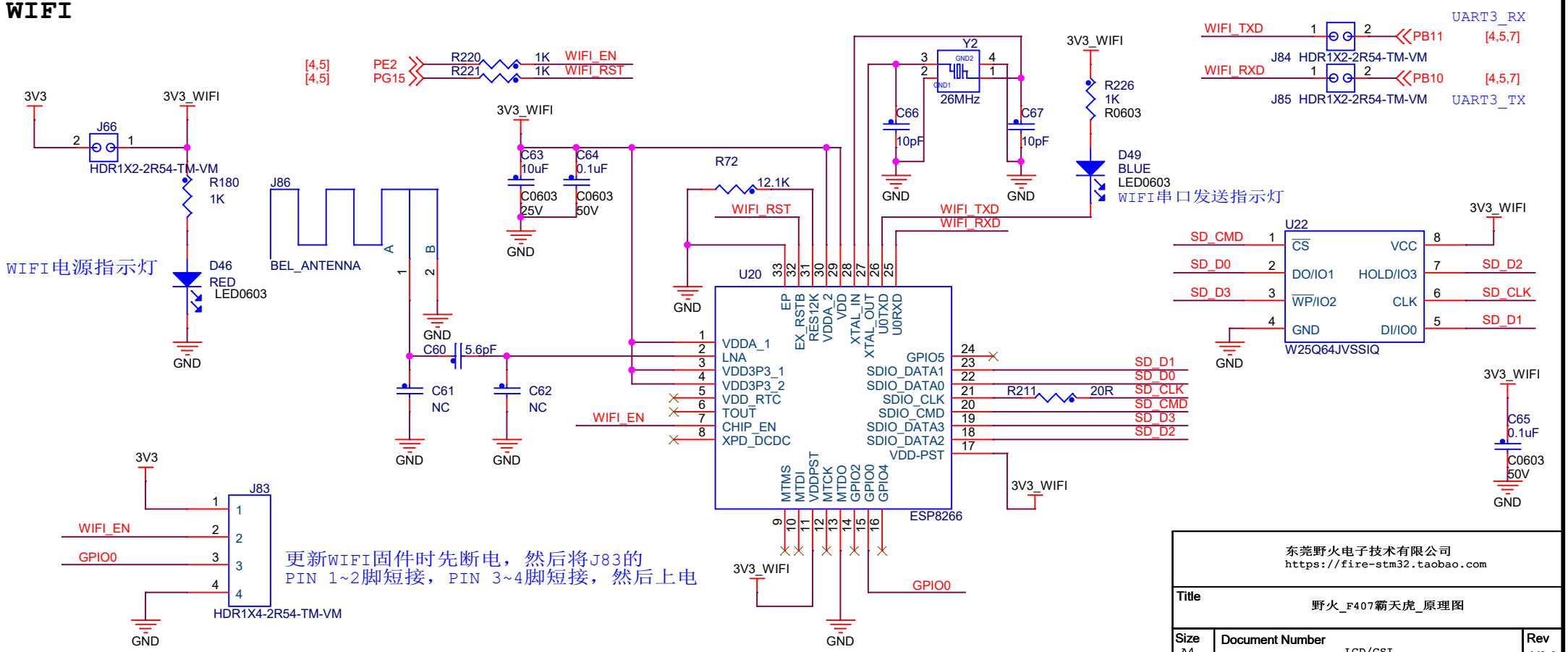
液晶接口-FSMC-8080模式



WIFI



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Title	野火_F407霸天虎_原理图
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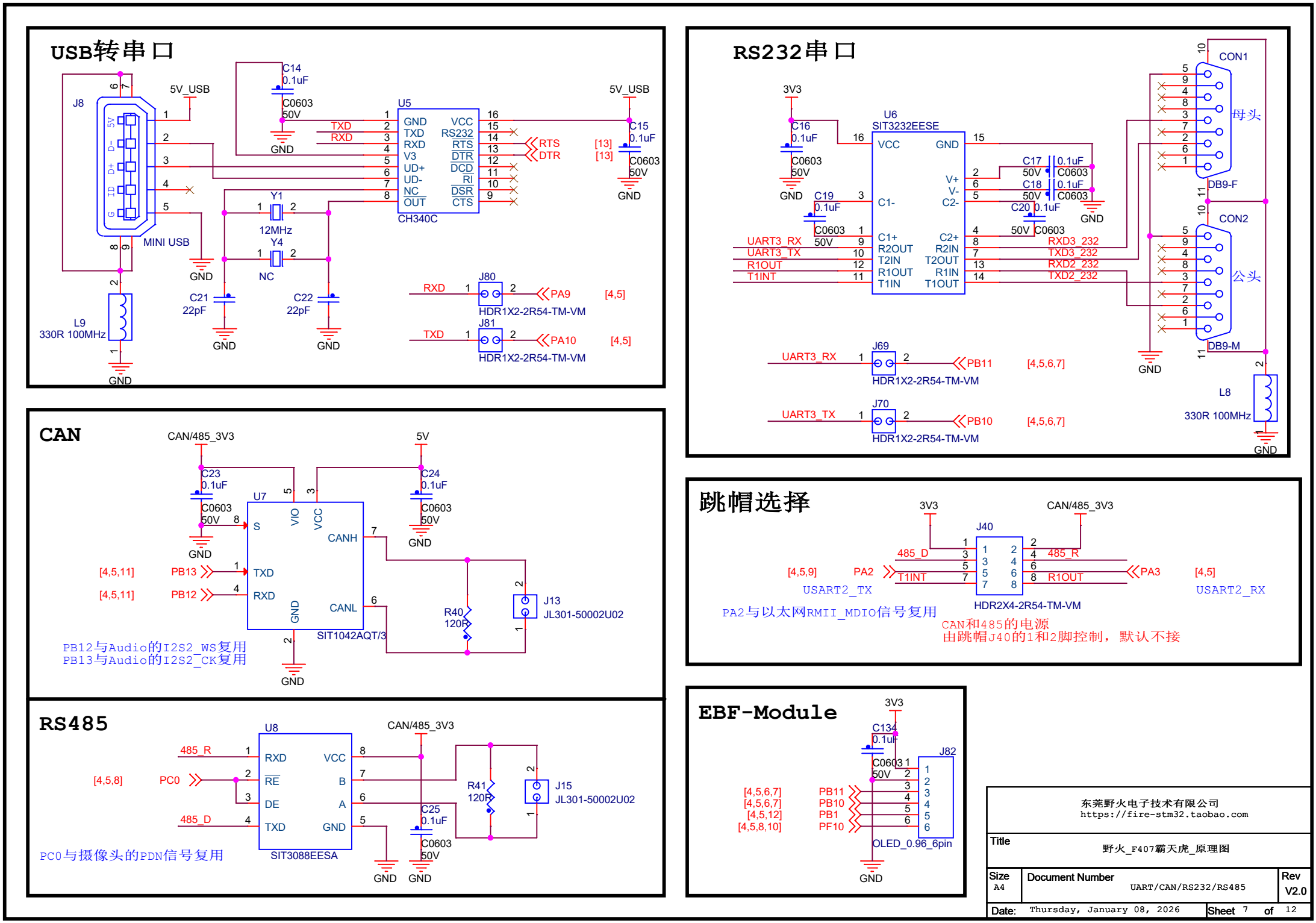
USB转串口
J8
MINI USB
5V_USB
C14
0.1uF
C0603
50V
GND
U5
CH340C
GND
TXD
RXD
V3
UD+
UD-
NC
OUT
VCC
RS232
RTS
DTR
DCD
RI
DSR
CTS
C15
0.1uF
[13]
[13]
Y1
12MHz
Y4
NC
C21
22pF
C22
22pF
L9
330R 100MHz
J80
HDR1X2-2R54-TM-VM
PA9
[4,5]
J81
PA10
[4,5]
RS232串口
U6
SIT3232EESE
3V3
C16
0.1uF
C19
0.1uF
C17
C18
C20
C1-
C1+
R2OUT
T2IN
R1OUT
T1IN
V+
V-
C2-
C2+
R2IN
T2OUT
R1IN
T1OUT
UART3_RX
UART3_TX
R1OUT
T1INT
RXD3_232
TXD3_232
RXD2_232
TXD2_232
CON1
母头
DB9-F
CON2
公头
DB9-M
J69
PB11
[4,5,6,7]
J70
PB10
[4,5,6,7]
L8
330R 100MHz
CAN
CAN/485_3V3
5V
C23
0.1uF
C24
0.1uF
U7
SIT1042AQT/3
S
VIO
VCC
CANH
CANL
TXD
RXD
GND
[4,5,11]
PB13
[4,5,11]
PB12
R40
120R
J13
JL301-50002U02
PB12与Audio的I2S2_WS复用
PB13与Audio的I2S2_CK复用
RS485
U8
SIT3088EESA
485_R
485_D
[4,5,8]
PC0
RE
DE
B
A
R41
120R
J15
JL301-50002U02
C25
0.1uF
PC0与摄像头的PDN信号复用
跳帽选择
J40
HDR2X4-2R54-TM-VM
[4,5,9]
PA2
USART2_TX
PA3
[4,5]
USART2_RX
PA2与以太网RMII_MDIO信号复用
CAN和485的电源
由跳帽J40的1和2脚控制，默认不接
EBF-Module
C134
0.1uF
J82
OLED_0.96_6pin
[4,5,6,7]
[4,5,6,7]
[4,5,12]
[4,5,8,10]
PB11
PB10
PB1
PF10
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Title
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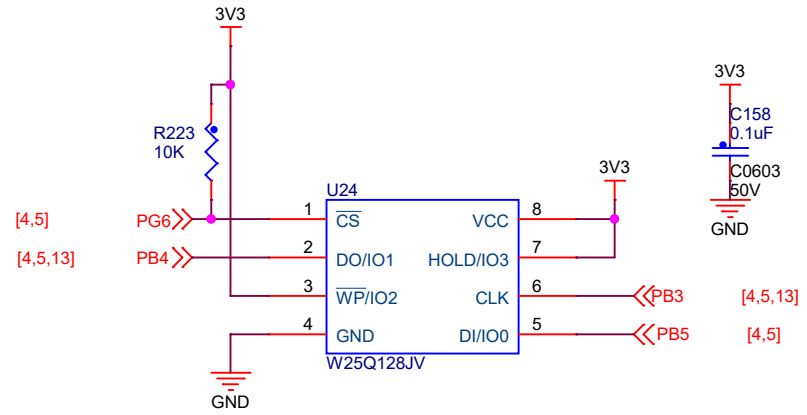
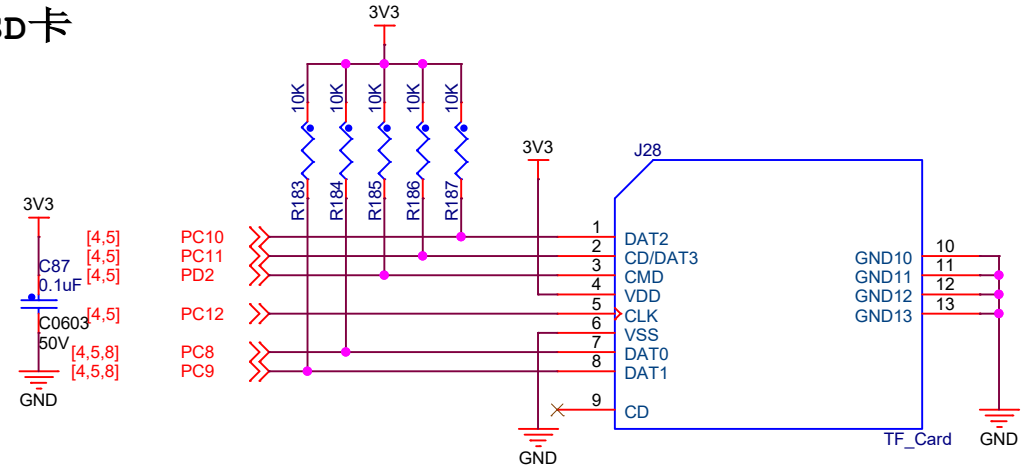
SPI_FLASH

容量：16M字节

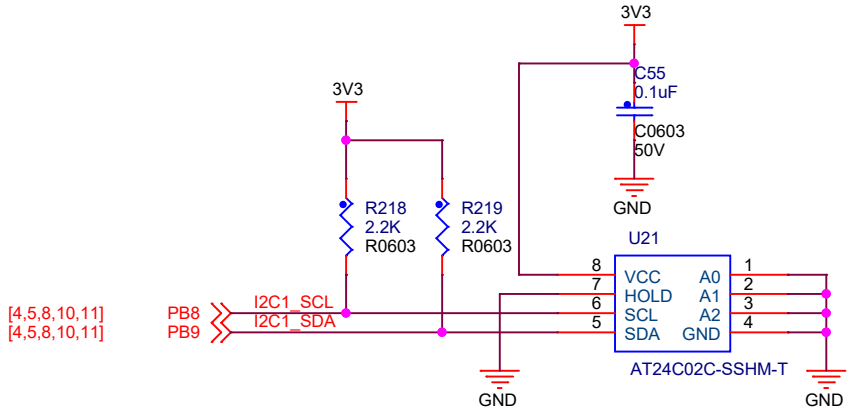


SD卡

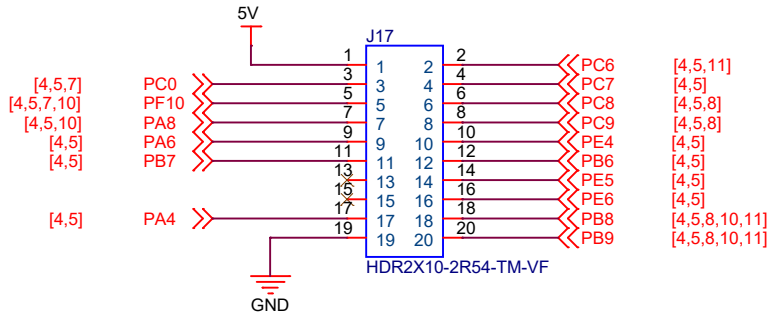


EEPROM

容量：256字节



摄像头接口



PC8与SDIO的SDIO_D0信号复用
PC9与SDIO的SDIO_D1信号复用
PF10与MPU6050的INT信号复用
PC0与RS485的RE信号复用

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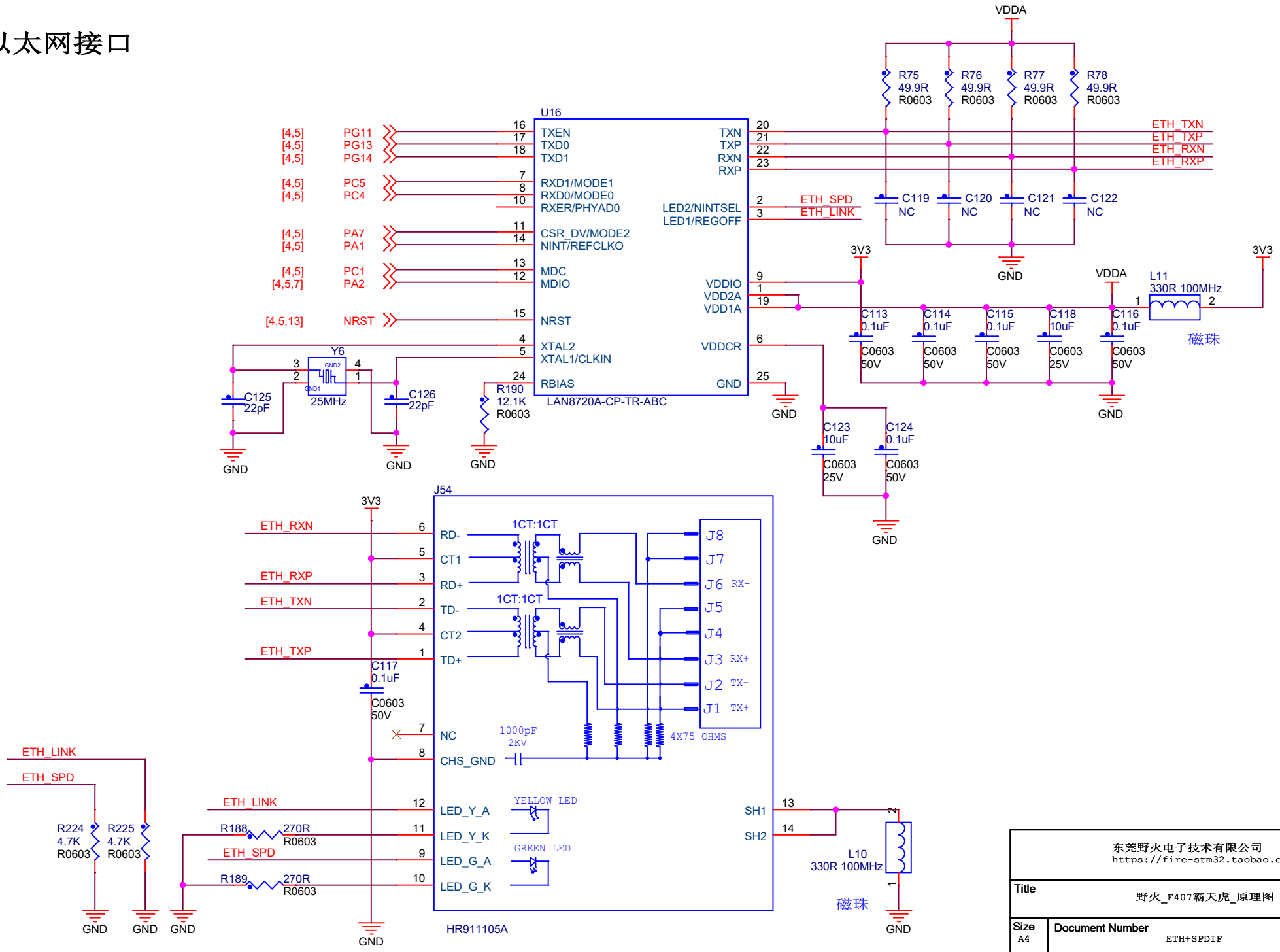
以太网接口



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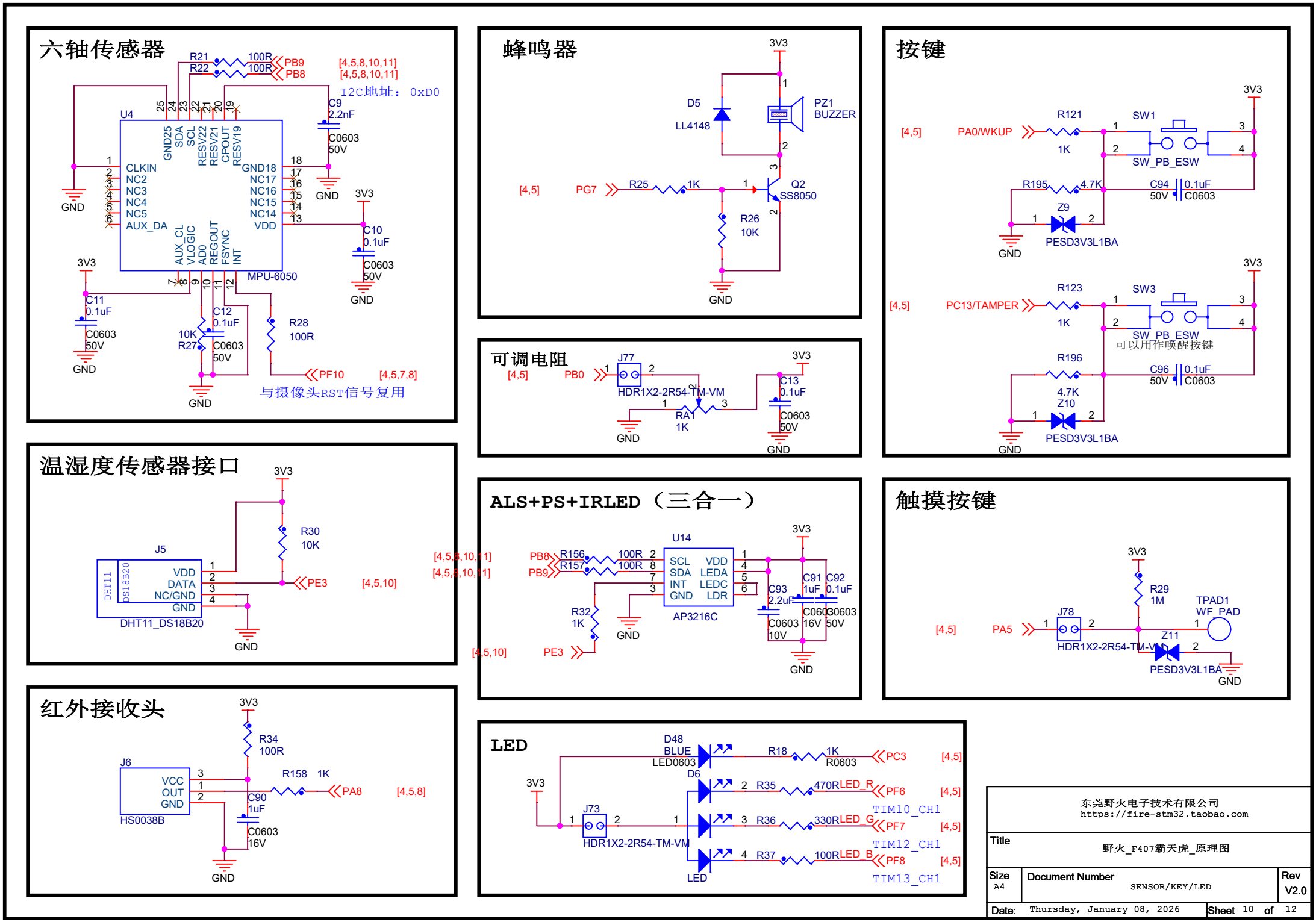
六轴传感器
U4
MPU-6050
CLKIN
NC2
NC3
NC4
NC5
AUX_DA
GND25
SDA
SCL
RESV22
RESV21
CPOUT
RESV19
GND18
NC17
NC16
NC15
NC14
VDD
AUX_CL
VLOGIC
AD0
REGOUT
FSYNC
INT
R21 100R PB9 [4,5,8,10,11]
R22 100R PB8 [4,5,8,10,11]
I2C地址： 0xD0
C9 2.2nF C0603 50V
C10 0.1uF C0603 50V
C11 0.1uF C0603 50V
C12 0.1uF C0603 50V
R27 10K
R28 100R
PF10 [4,5,7,8]
与摄像头RST信号复用
3V3
GND
蜂鸣器
D5 LL4148
PZ1 BUZZER
[4,5] PG7 R25 1K
Q2 SS8050
R26 10K
按键
[4,5] PA0/WKUP R121 1K
SW1 SW_PB_ESW
R195 4.7K
C94 0.1uF 50V C0603
Z9 PESD3V3L1BA
[4,5] PC13/TAMPER R123 1K
SW3 SW_PB_ESW
可以用作唤醒按键
R196 4.7K
C96 0.1uF 50V C0603
Z10 PESD3V3L1BA
可调电阻
[4,5] PB0 J77 HDR1X2-2R54-TM-VM
RA1 1K
C13 0.1uF C0603 50V
温湿度传感器接口
J5 DHT11_DS18B20
VDD
DATA
NC/GND
GND
R30 10K
PE3 [4,5,10]
ALS+PS+IRLED（三合一）
U14 AP3216C
SCL
SDA
INT
GND
VDD
LEDA
LEDC
LDR
PB8 R156 100R [4,5,8,10,11]
PB9 R157 100R [4,5,8,10,11]
R32 1K
[4,5,10] PE3
C93 2.2uF C0603 10V
C91 1uF C0603 16V
C92 0.1uF C0603 50V
触摸按键
R29 1M
TPAD1 WF_PAD
[4,5] PA5 J78 HDR1X2-2R54-TM-VM
Z11 PESD3V3L1BA
红外接收头
J6 HS0038B
VCC
OUT
GND
R34 100R
R158 1K PA8 [4,5,8]
C90 1uF C0603 16V
LED
D48 BLUE LED0603
D6
R18 1K R0603 PC3 [4,5]
R35 470R LED_R PF6 [4,5] TIM10_CH1
R36 330R LED_G PF7 [4,5] TIM12_CH1
R37 100R LED_B PF8 [4,5] TIM13_CH1
J73 HDR1X2-2R54-TM-VM
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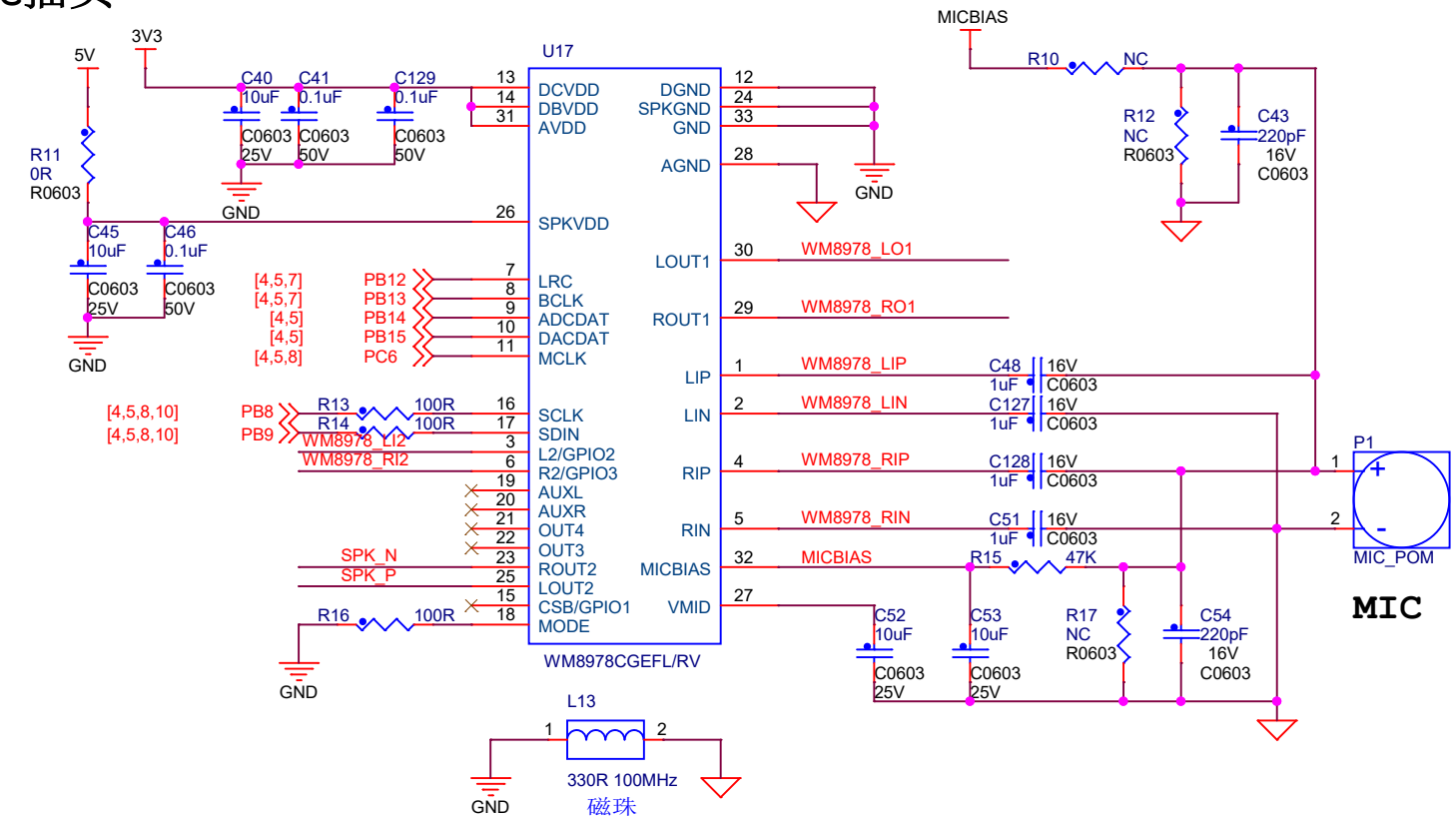
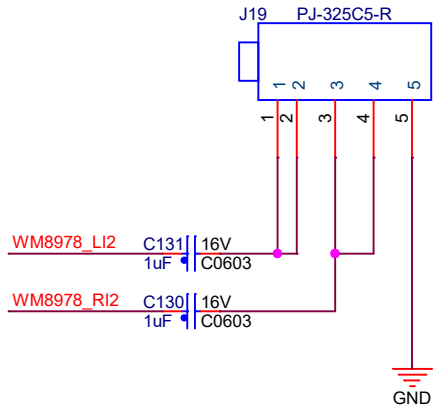
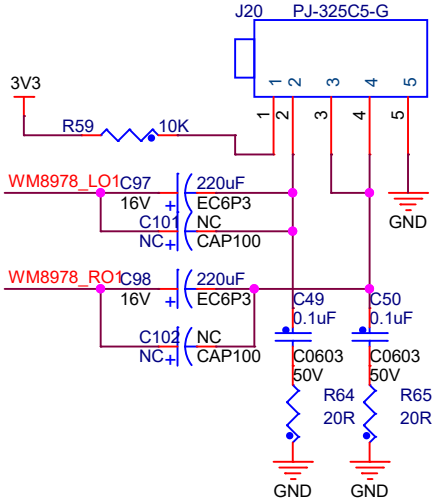
MIC插头



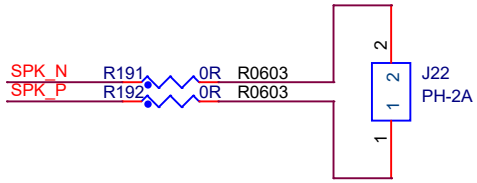
音频输入



耳机插座



喇叭插座



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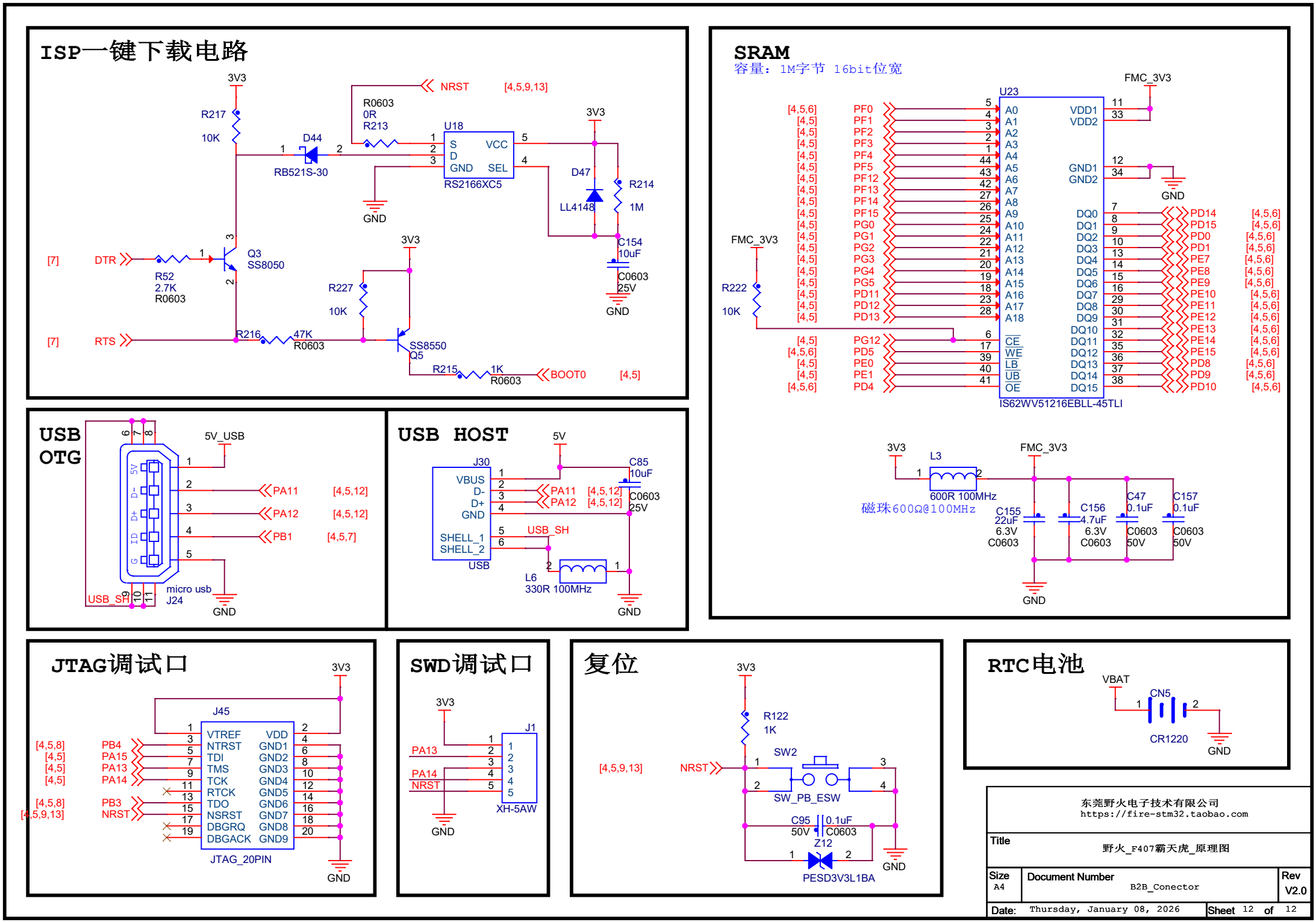
ISP一键下载电路
3V3
R217
10K
D44
RB521S-30
R0603
0R
R213
NRST
[4,5,9,13]
U18
S
D
GND
VCC
SEL
RS2166XC5
D47
LL4148
R214
1M
C154
10uF
C0603
25V
GND
[7]
DTR
R52
2.7K
R0603
Q3
SS8050
R227
10K
R216
47K
R0603
[7]
RTS
SS8550
Q5
R215
1K
R0603
BOOT0
[4,5]
SRAM
容量：1M字节 16bit位宽
FMC_3V3
U23
A0
A1
A2
A3
A4
A5
A6
A7
A8
A9
A10
A11
A12
A13
A14
A15
A16
A17
A18
CE
WE
LB
UB
OE
VDD1
VDD2
GND1
GND2
DQ0
DQ1
DQ2
DQ3
DQ4
DQ5
DQ6
DQ7
DQ8
DQ9
DQ10
DQ11
DQ12
DQ13
DQ14
DQ15
PF0
PF1
PF2
PF3
PF4
PF5
PF12
PF13
PF14
PF15
PG0
PG1
PG2
PG3
PG4
PG5
PD11
PD12
PD13
PG12
PD5
PE0
PE1
PD4
PD14
PD15
PD0
PD1
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD8
PD9
PD10
R222
10K
IS62WV51216EBLL-45TLI
L3
600R 100MHz
磁珠600Ω@100MHz
C155
22uF
6.3V
C0603
C156
4.7uF
6.3V
C0603
C47
0.1uF
C0603
50V
C157
0.1uF
C0603
50V
USB OTG
5V_USB
PA11
PA12
PB1
USB_SH
micro usb
J24
USB HOST
J30
VBUS
D-
D+
GND
SHELL_1
SHELL_2
USB
C85
10uF
C0603
25V
USB_SH
L6
330R 100MHz
JTAG调试口
J45
VTREF
NTRST
TDI
TMS
TCK
RTCK
TDO
NSRST
DBGRQ
DBGACK
VDD
GND1
GND2
GND3
GND4
GND5
GND6
GND7
GND8
GND9
PB4
PA15
PA13
PA14
PB3
NRST
JTAG_20PIN
SWD调试口
J1
XH-5AW
复位
R122
1K
SW2
SW_PB_ESW
C95
0.1uF
50V
C0603
Z12
PESD3V3L1BA
RTC电池
VBAT
CN5
CR1220
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